Journal of Molecular Spectroscopy 368 (2020) 111256

Contents lists available at ScienceDirect

Journal of Molecular Spectroscopy

journal homepage: www.elsevier.com/locate/jms

Structure of the model grignard-type reagent $ClZnCH_3$ ($\tilde{X}^1A_1$) by millimeter-wave spectroscopy

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ARTICLE INFO

Article history:
Received 7 December 2019
In revised form 24 January 2020
Accepted 27 January 2020
Available online 29 January 2020

Keywords:
Millimeter-wave spectroscopy
$ClZnCH_3$
Organozinc compounds
Molecular structure

ABSTRACT

Pure rotational spectra of the $^{37}ClZnCH_3$, $ClZnCD_3$, and $ClZn^{13}CH_3$ isotopologues of monomeric $ClZnCH_3$ ($\tilde{X}^1A_1$) have been recorded using millimeter-wave direct absorption techniques in the frequency range 263–303 GHz. These species were synthesized in the gas phase in a DC discharge by the reaction of zinc vapor, produced in a Broida-type oven, with $^{37}ClCH_3$ (in natural chlorine abundance), $ClCD_3$, or $Cl^{13}CH_3$. Five to eight rotational transitions $J + 1 \leftarrow J$ consisting of K ladder structure were measured for each isotopologue, identifying all three species as prolate symmetric tops. The data for each isotopologue were analyzed with a symmetric top Hamiltonian and rotational and centrifugal constants determined. In combination with previous measurements of $Cl^{64}ZnCH_3$, $Cl^{66}ZnCH_3$, and $Cl^{68}ZnCH_3$, an $r_m^{(2)}$ structure was determined for this organozinc compound. The bond lengths in $ClZnCH_3$ were calculated to be $r_{Cl\text{-}Zn} = 2.0831(1)$ Å, $r_{Zn\text{-}C} = 1.9085(1)$ Å, and $r_{C\text{-}H} = 1.1806(5)$ Å, considerably different from those established from crystal structures of related species $ClZnCH_2CH_3$ and ClZnEtTMEDA. The H—C—H bond angle was found to be 110.5° – slightly larger than that in methane. These data serve to benchmark future structure calculations of organozinc compounds, which are widely used in organic synthesis.

1. Introduction

Organozinc compounds RZnX continue to be prominent in organic synthesis, particularly in the formation of carbon-carbon bonds [1]. The Barbier-Grignard-type reaction, for example, which involves zinc and an organic halide, has been developed to produce homoallylic alcohols, as well as α, β-unsaturated carbonyl compounds, nitriles, pyrroles, and indoles [2,3]. In the Negishi cross-coupling reaction, the organozinc reagent R'ZnX plays the critical role of transferring the R′ group (usually alkenyl, aryl, allyl, alkyl, or benzyl) to the palladium or nickel catalyst to form the R-R' product [4–6]. Recently, the nucleophilic addition of an organozinc reagent such as $BrZnRCO_2Et$ to an aryldiazonium salt has been used as an alternative to the classic Fischer indole synthesis, allowing for regioselectivity in the formation of indole-containing natural products [7,8].

Because of their intrinsic versatility in synthesis, theoretical studies have been conducted of organozinc compounds to better elucidate their chemical properties and the energetics of their associated reactions. Lin and Phillips, for example, employed DFT methods to examine Negishi alkyl-alkyl cross-coupling using CH_3ZnI as the model alkyl nucleophile [9]. Also using DFT, de Jong et al. studied the activation of C—H and C—Cl bonds by various metals, including Zn [10]. Both works calculated the structures of CH_3ZnX compounds (X = I, Cl), and found them to have C_{3v} symmetry with a linear X—Zn—C backbone.

Experimental structural studies of these compounds are naturally of interest. As early as 1996, X-ray crystallography has been conducted to produce geometric information for organozinc species. Early work was carried out by Moseley and Shearer, who measured the structure of solid ethylzinc iodide [11,12]. They found that the basic orthorhombic cell consisted of four EtZnI units with two different Zn—C bond lengths. Shortly thereafter, Andrews at al. [13] synthesized and measured the crystal structure of the more complex ClZnEtTMEDA, finding the compound to possess a distorted tetrahedral metal center. More recently, X-ray crystallography has generated structures of EtZnCl [14] and $MesBipyCH_2(ZnI)_2$ [15]. Unlike the iodide analog, ethylzinc chloride was found to consist of puckered sheets with Zn_3Cl_3 rings, with each zinc tetrahedrally-coordinated to one ethyl group and three chlorine atoms. The di(iodozincio)methane complex was found to have two zinc centers bridged by a methylene group, stabilized by two bulky 6-mesityl-2,2-bipyridine (MesBipy) ligands.

Recently, more precise structures of simple organozinc compounds have been determined using gas-phase rotational

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https://doi.org/10.1016/j.jms.2020.111256
0022-2852/

Table 1
Observed Transition Frequencies for $ClZnCH_3$ ($\tilde{X}^1A_1$).[a]

$J' \leftarrow J''$	K	$ClZn^{13}CH_3$		$^{37}ClZnCH_3$		$ClZnCD_3$	
		ν_{obs}	$\nu_o - \nu_c$	ν_{obs}	$\nu_o - \nu_c$	ν_{obs}	$\nu_o - \nu_c$
53 ← 52	0			263301.778	−0.013		
	1			263297.730	0.020		
	3			263265.053	−0.014		
	4			263236.503	−0.002		
	5			263199.799	0.018		
54 ← 53	1			268256.093	0.027		
	4			268193.700	−0.041		
	5			268156.336	−0.010		
55 ← 54	0			273218.184	0.067		
	1			273213.913	0.026		
	2			273201.297	0.098		
	3			273180.020	−0.032		
	4			273150.375	−0.070		
	5			273112.456	0.076		
	6			273065.903	0.047		
56 ← 55	0			278175.383	−0.086		
	1			278171.174	0.009		
	2			278158.178	−0.075		
	3			278136.760	0.026		
	4			278106.544	−0.063		
	5			278067.927	0.054		
	6			278020.500	−0.031		
57 ← 56	0	–	–	283132.328	0.062		
	1	283847.445	0.002	283127.790	−0.098		
	2	283834.483	0.040	283114.655	−0.099		
	3	283812.761	−0.013	283092.854	−0.010		
	4	283782.403	−0.036	283062.145	−0.073		
	5	283743.418	−0.018	283022.875	0.059		
	6	283695.768	0.002	282974.639	−0.018		
58 ← 57	0	288820.743	0.021	288088.608	0.108		
	1	288816.311	−0.004	–	–		
	2	288803.076	−0.016	288070.747	0.055		
	3	288781.033	−0.022	288048.441	0.010		
	4	288750.170	−0.033	–	–		
	5	288710.541	0.005	287977.127	−0.071		
	6	288662.052	−0.003	287928.205	−0.019		
59 ← 58	0	293789.146	0.048	293044.148	−0.012		
	1	293784.623	0.007	293039.626	−0.008		
	2	293771.134	−0.039	293026.077	0.021		
	3	293748.749	−0.019	293003.454	0.028		
	4	293717.353	−0.047	292971.789	0.046		
	5	293677.124	0.054	–	–		
	6	293627.847	0.069	–	–		
60 ← 59	0	298756.978	0.084	297999.227	−0.009	275190.867	−0.010
	1	298752.348	0.009	–	–	275187.506	−0.024
	2	298738.662	−0.013	297980.900	0.063	275177.487	−0.002
	3	298715.887	−0.015	297957.766	−0.072	275160.738	−0.015
	4	298684.045	0.025	297925.583	−0.056	275137.287	−0.037
	5	298643.042	0.014	297884.250	0.009	275107.153	−0.047
	6	298592.919	−0.008	297833.696	0.053	275070.376	−0.006
61 ← 60	0	303724.124	0.022			279767.590	−0.007
	1	303719.454	−0.020			279764.341	0.144
	2	303705.596	0.007			–	–
	3	303682.422	−0.027			279736.992	0.002
	4	303650.015	−0.038			279713.134	−0.050
	5	303608.324	−0.076			279682.535	−0.041
	6	303557.530	0.039			279645.307	0.141
62 ← 61	0					284343.859	0.027
	1					284340.368	−0.010
	2					284330.012	−0.002
	3					284312.734	−0.007
	4					284288.602	0.043
	5					284257.417	−0.051
	6					284219.482	0.013
63 ← 62	0					288919.629	0.056
	1					288916.027	−0.038
	2					288905.529	−0.012
	3					288887.997	−0.003
	4					288863.429	−0.014
	5					288831.866	−0.004
	6					288793.256	−0.025

Table 1 (*continued*)

$J' \leftarrow J''$	K	$ClZn^{13}CH_3$		$^{37}ClZnCH_3$		$ClZnCD_3$	
		ν_{obs}	$\nu_o - \nu_c$	ν_{obs}	$\nu_o - \nu_c$	ν_{obs}	$\nu_o - \nu_c$
64 ← 63	0					293494.862	0.050
	1					293491.251	0.000
	2					293480.545	−0.021
	3					293462.676	−0.082
	4					293437.889	0.062
	5					293405.664	−0.108
	6					293366.673	0.078

[a] All values in MHz.

spectroscopy. In 2010, Flory et al. [16] measured both the millimeter-wave and the Fourier transform microwave (FTMW) spectrum of $HZnCH_3$. Through multiple isotopic substitution, these authors determined an accurate r_0 structure for the molecule in its monomeric form, which clearly had C_{3v} symmetry and a Zn—C bond length of 1.928 Å. This work was followed by a similar purely millimeter-wave investigation of $IZnCH_3$ and its zinc and deuterium isotopologues [17]; a C_{3v} geometry was found for this species as well, with an almost identical Zn-C bond distance of 1.920 Å. A subsequent study of $ClZnCH_3$ was carried out using both millimeter-wave and FTMW methods [18]. The measured spectrum, again, was that of a symmetric top, indicating C_{3v} symmetry in this molecule. However, only the zinc atom was isotopically substituted, which was not adequate to derive an accurate structure for comparison with the iodine analog.

In order to complete our structural analysis of $ClZnCH_3$, and make a more direct comparison with theory, we have measured millimeter-wave spectra of three new isotopologues of this molecule: $^{37}ClZnCH_3$, $ClZnCD_3$, and $ClZn^{13}CH_3$. Multiple rotational transitions in the range 263–303 GHz were recorded for each species, each exhibiting a classic prolate symmetric-top pattern. Combined with our previous study of the zinc isotopologues of $ClZnCH_3$, a more accurate $r_m^{(2)}$ structure has been determined for this model organozinc compound in its monomeric form. In this paper, we present our additional isotopologue measurements for the $ClZnCH_3$ species and compare our structural results with theory and other experimental methyl zinc compounds.

2. Experimental

The pure rotational spectra of $^{37}ClZnCH_3$, $ClZnCD_3$, and $ClZn^{13}CH_3$ were recorded using the low temperature millimeter-wave direct absorption spectrometer of the Ziurys group [19]. The system consists of three basic units: a radiation source, a reaction cell, and a detector. Millimeter radiation in the range of 65–142 GHz is produced by Gunn oscillators, and is subsequently doubled, tripled, quadrupled, etc. to higher frequencies (~150–850 GHz) using Schottky diode multipliers. The radiation is guided into the stainless-steel reaction cell (~2 ft long, 0.5 ft in diameter) using Teflon lenses and a polarizing grid. At the back of the cell is a rooftop mirror, which reflects the radiation back through the cell for a second pass after rotating its polarization by 90°. Also attached to the cell is a Broida-type oven, which allows for continuous heating of pure metal. The radiation then emerges from the cell and is reflected off the grid into the detector, a helium-cooled InSb hot electron bolometer. Source modulation at a rate of 25 kHz allows for phase-sensitive detection; signals are demodulated at 2*f* by a lock-in amplifier to produce a second-derivative line shape.

The $ClZnCH_3$ isotopologues were synthesized in a DC discharge by reacting melted zinc, generated by the Broida-type oven, with $ClCH_3$ or an isotopic variant. The $^{37}ClZnCH_3$ species was studied in natural abundance of chlorine (^{35}Cl:^{37}Cl = 3:1), whereas $ClCD_3$ and $Cl^{13}CH_3$ were used to generate $ClZnCD_3$ and $ClZn^{13}CH_3$. Optimized signals were produced when ~10 mTorr of the respective reactant gas was introduced above the oven. Additionally, argon was flowed from beneath the oven (~20 mTorr) to entrain metal vapor, as well as around the Teflon lenses (~25 mTorr) to prevent coating by metal vapor. A 900 mA (170 V) DC discharge was found to be optimal in creating the $ClZnCH_3$ variants, presumably by the insertion of zinc into methyl chloride, as indicated by our previous experiments [16–18].

Transition frequencies were determined by averaging two scans, 5 MHz in total range centered on a given spectral line, one ascending and the other descending in frequency. Each observed line was fit with a Gaussian profile to determine the transition frequency. The experimental uncertainty is estimated to be ±50 kHz.

3. Results and analysis

The initial search for the spectra of $^{37}ClZnCH_3$, $ClZn^{13}CH_3$, and $ClZnCD_3$ was guided by mass scaling of the rotational constant from those previously reported for the zinc isotopologues of $ClZnCH_3$ [18]. Based on those values, and the centrifugal distortion constants for $ClZnCH_3$, spectra were predicted and the characteristic symmetric top patterns sought. Because both the ^{37}Cl and ^{13}C isotopic substitutions lie along the C_{3v} symmetry axis, the expected patterns were found within ~50 MHz of the predicted values. Once one symmetric top pattern was found, additional transitions were readily located for both species. In the case of $ClZnCD_3$, however, the mass change is off-axis, resulting in a greater deviation from the predicted rotational constant. To locate a believable harmonic pattern, ~10 GHz in frequency was continuously scanned before the species was definitely found. This wider search was often hampered by contaminating transitions of $ClCH_3$ and its vibrational satellite lines, which also exhibit a symmetric top pattern with similar *K*-component splittings.

Table 1 presents the transitions measured for $^{37}ClZnCH_3$, $ClZn^{13}CH_3$, and $ClZnCD_3$. As the table shows, between 5 and 8 rotational transitions for $J + 1 \leftarrow J$ were recorded for each species, each consisting of 3–7 *K* components, resulting in a total of 113 lines. Not all *K* components of a given transition could be measured because of contamination from the precursor gas. All observed rotational transitions ($J = 53 \leftarrow 52$ to $J = 64 \leftarrow 63$) exhibit decreasing values of *K* with increasing frequency, indicative of a prolate symmetric top with $D_{JK} > 0$.

Fig. 1 shows representative spectra of the three studied $ClZnCH_3$ isotopologues. In the top panel, the $J = 54 \leftarrow 53$ rotational transition of $^{37}ClZnCH_3$ near 273 GHz is presented, consisting of the K = 0–4 components. The middle panel shows the $J = 62 \leftarrow 61$ transition of $ClZnCD_3$ near 284 GHz; the K = 0 through 5 lines are visible in this spectrum. The $J = 58 \leftarrow 57$ transition of $ClZn^{13}CH_3$ near 289 GHz is displayed in the bottom panel, with K = 0–4

Table 2
Spectroscopic Constants for $ClZnCH_3$ ($\tilde{X}^1A_1$).[a]

Parameter	$ClZn^{13}CH_3$	$^{37}ClZnCH_3$	$ClZnCD_3$	$Cl^{64}ZnCH_3$ [b]	$Cl^{66}ZnCH_3$ [b]	$Cl^{68}ZnCH_3$ [b]
B	2492.5853(70)	2486.2952(38)	2295.6459(66)	2558.39022(56)	2554.76481(66)	2551.26855(61)
$D_J \times 10^4$	4.0895(99)	4.1226(59)	3.3174(85)	4.3354(10)	4.3267(12)	4.3176(14)
D_{JK}	0.03856(38)	0.03907(23)	0.02838(36)	0.04081(20)	0.04053(24)	0.040388(29)
$H_{JK} \times 10^8$	8.4(5.3)	10.3(3.6)	6.8(4.7)	8.9(3.3)	4.6(3.9)	
eQq (Cl)				−27.443(41)	−27.89(31)	−28.14(52)
rms	0.034	0.053	0.052	0.066	0.079	0.081

[a] All values in MHz; errors are 3σ.
[b] From Ref. [18].

Table 3
Structural Parameters for $ClZnCH_3$ ($\tilde{X}^1A_1$) and Related Species.

Species	r_{X-Zn} (Å)	r_{Zn-C} (Å)	r_{C-H} (Å)	$\angle_{Zn-C-H}$ (°)	$\angle_{H-C-H}$ (°)	Method	Ref
$ClZnCH_3$	2.088(4)	1.912(7)	1.1(5)	109(16)	110(8)	r_0	This work
	2.0831(1)	1.9085(1)	1.1806(5)	108.4(1)	110.5(1)	$r_m^{(2)}$ [a]	This work
	2.129	1.937	1.096	109.6		Theory, r_e	[9]
$IZnCH_3$	2.4076(3)	1.9202(9)	1.180[b]	108.2(4)	110.7(1)	r_0	[17]
ZnCl	2.1300(1)					r_e	[25]
	2.221					Theory, r_e	[26]
$ClZnCH_2CH_3$	2.3486(6) [c]	1.946(2)				Crystal	[14]
ClZnEtTMEDA	2.269(3)	1.94(1)				Crystal	[13]

[a] c_b = 0.117(3); d_b = -0.58(2).
[b] Held fixed.
[c] Longer bond lengths of 2.5248(6) Å and 2.5408(7) Å also reported.

components present. In all spectra, asterisks denote unknown or precursor features. The classic symmetric top pattern is visible in all data, with the $K = 3$ line typically stronger than the others, reflecting the proton or deuteron nuclear spin statistics of the methyl group [20]. Note, for the hydrogen and deuterium species respectively, the $K = 3n$ components should be a factor of 2 and 1.375 larger than the $K \neq 3n$ components.

Each isotopologue was fit separately with a symmetric top Hamiltonian using the non-linear least squares fitting program SPFIT [21]. For a symmetric top molecule, the rotational levels are modeled with the following energy expression [20]:

$$F(J, K) = BJ(J+1) - D_J(J(J+1))^2 + (A-B)K^2 - D_{JK}J(J+1)K^2 - D_K K^4. \quad (1)$$

In the analysis, the A rotational and D_K centrifugal distortion constants could not be determined as only $\Delta K = 0$ transitions were observed, and were therefore set to zero. All other constants were determined in the fits (along with H_{JK}); the results of the analyses for the three isotopologues $^{37}ClZnCH_3$, $ClZn^{13}CH_3$, and $ClZnCD_3$ are summarized in Table 2. Also included in the table are the parameters previously measured for $Cl^{64}ZnCH_3$, $Cl^{66}ZnCH_3$, and $Cl^{68}ZnCH_3$ from Ref. [18]. The new spectroscopic constants are consistent with those previously measured. As shown in the table, the rms of the three fits is < 54 kHz – on par with the experimental uncertainty.

With these additional isotopically-substituted species, and the previous zinc isotopologues, both r_0 and $r_m^{(2)}$ structures were determined for $ClZnCH_3$. A nonlinear least-squares analysis of the six rotational constants in Table 2 was employed to establish the geometry using the program STRFIT [22]. Because of the numerous isotopic substitutions for $ClZnCH_3$, an $r_m^{(2)}$ could be calculated, which takes into account the isotopic dependence of zero-point energy differences [23]. Therefore, it is more comparable to the equilibrium structure (r_e) of the molecule. The resulting structure of $ClZnCH_3$ and related species are summarized in Table 3.

4. Discussion

The new measurements of three additional isotopologues of $ClZnCH_3$ are further evidence for the C_{3v} symmetry and $^1A_1$ ground state assignment of this model organozinc compound. They additionally verify the formation of the molecule by zinc insertion into the Cl—C bond. As previously discussed [16], the DC discharge used in the synthesis appears to activate the atomic zinc, rather than simply create radical species. These measurements also enabled a more reliable determination of the structure of this species for comparison with theory and X-ray crystallography data. In the previous study of $ClZnCH_3$ [18] only the zinc atom was substituted. Consequently, only the Cl-Zn bond distance (2.23(7) Å) could be calculated from the rotational constants, and all the other structural parameters were fixed to that of $IZnCH_3$.

As shown in Table 3 and in Fig. 2, the bond lengths in $ClZnCH_3$ are r_{Cl-Zn} = 2.0831(1) Å, r_{Zn-C} = 1.9085(1) Å, and r_{C-H} = 1.1806(5) Å. The Zn—C—H and H—C—H angles are estimated to be 108.4(1)° and 110.5(1)°, respectively. Note that the r_0 structure has considerably more uncertainty because the deuterium substitution is not properly accounted for. In comparison, the Cl—Zn bond length determined in the previous study was 2.23 Å – larger by ~0.14 Å but understandable given the assumptions. In contrast, the r_e geometry predicted by theory is r_{Cl-Zn} = 2.129 Å, r_{Zn-C} = 1.937 Å, and r_{C-H} = 1.096 Å with a Zn—C—H angle of 109.6° [9]. The calculated Cl—Zn and Zn—C bond distances are ~0.03–0.04 Å longer than those determined experimentally, and the Zn—C—H angle about a degree larger. The computed C—H bond length is ~0.08 Å shorter. The computational results are therefore reasonably close to the experimental values, but could be further refined.

The bond lengths determined from X-ray crystallography for the larger $ClZnCH_2CH_3$ and ClZnEtTMEDA species are significantly longer than for those in $ClZnCH_3$. The Cl—Zn bond distances in the X-ray data are 2.3486(6) Å and 2.269(3) Å for the ethyl and TMEDA compounds, respectively [13,14] – approximately 0.2–0.3 Å longer than in $ClZnCH_3$. The Zn—C bond distances are closer in value (1.946(2) Å and 1.94(1) Å), with a difference of ~0.03–0.04 Å. Each zinc atom in $ClZnCH_2CH_3$ and ClZnEtTMEDA is tetravalent, causing steric effects that are likely increasing the bond lengths relative to the metal center.

A structural comparison can also be made with the analog halide species, $IZnCH_3$ [17]. The iodine atom cannot be substituted, due to there only being one isotope; therefore, the C—H bond distance was held fixed to that determined here (1.180 Å). As might

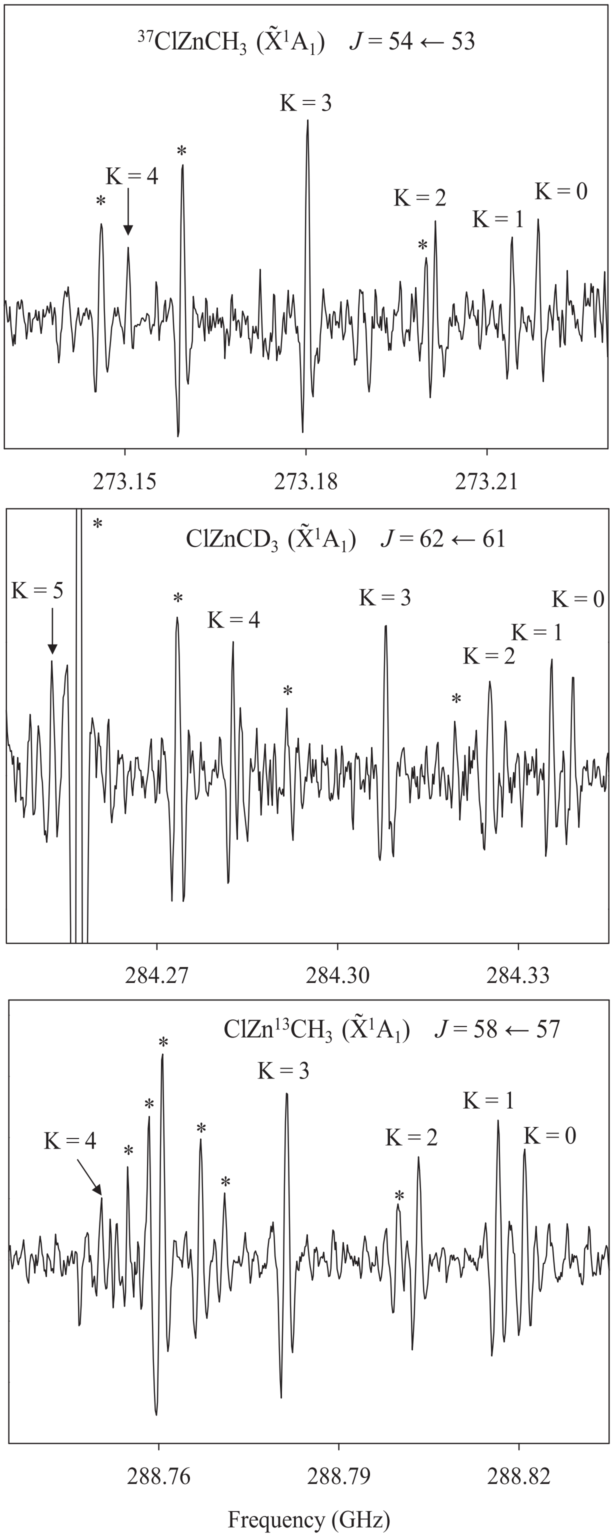

Fig. 1. Representative millimeter-wave spectra of ^{37}ClZnCH$_3$ (top panel), ClZnCD$_3$ (middle panel), and ClZn^{13}CH$_3$ (bottom panel), showing the $J = 54 \leftarrow 53$, $J = 62 \leftarrow 61$, and $J = 58 \leftarrow 57$ rotational transitions near 273, 284, and 289 GHz respectively. Each spectrum consists of multiple K components, labeled on the figure, with the approximate relative spacing of 1:3:5:7 for $K = 0, 1, 2, 3, 4, 5$ – a classic symmetric top pattern for a $\tilde{X}^1A_1$ state. The $K = 3$ line is noticeably more intense than the other components for ^{37}ClZnCH$_3$ and ClZn^{13}CH$_3$, as expected for fermion statistics. Asterisks denote contamination due to the reactant gas and other possible species. Each spectrum was created from a 110 MHz scan, taken in ~70 s.

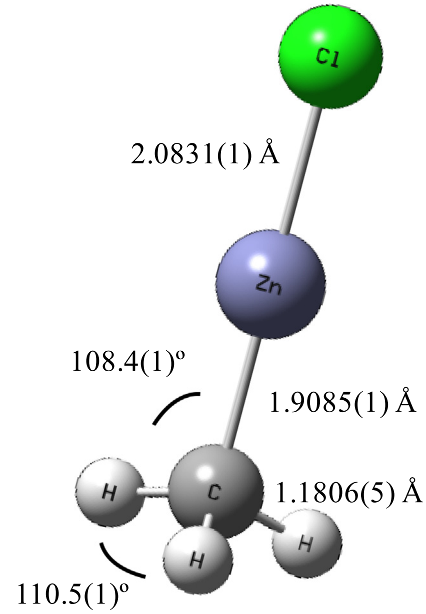

Fig. 2. Experimentally-determined $r_m^{(2)}$ structure for ClZnCH$_3$ ($\tilde{X}^1A_1$), established from the rotational constants of ^{37}ClZnCH$_3$, ClZnCD$_3$, and ClZn^{13}CH$_3$ measured here and those for Cl^{64}ZnCH$_3$, Cl^{66}ZnCH$_3$, and Cl^{68}ZnCH$_3$ from Ref. [18].

be expected, the Zn—X bond length increases with the size of the halide atom, over 0.3 Å from chlorine to iodine (see Table 3). There is less of an effect for the Zn—C bond length, which increases very slightly from Cl to I, ~0.01 Å. Because the C—H bond distance was fixed, it is problematic, however, to compare the H—C—H bond angle between these two species. Interestingly, the H—C—H bond angle in methane is 109.5° – smaller by about a degree than in ClZnCH$_3$ [24]. The C—H bond length in methane is 1.094 Å, ~0.09 Å shorter than this bond in ClZnCH$_3$. The substitution of hydrogen with XZn appears to distort the tetrahedral structure of methane, slightly opening the H—C—H bond angle, but the deviation is relatively small.

It is also of note that the Zn—Cl bond length is shorter for ClZnCH$_3$ by ~0.04 Å than for the ZnCl radical, which has one unpaired electron [25]. Filling the open electron shell has an obvious stabilizing effect on ClZnCH$_3$, suggesting some covalency in the bonding of the molecule. This conclusion was also deduced from the analysis of the chlorine electric quadrupole constant, as measured by the previous work [18].

5. Conclusion

Measurements of the rotational spectra of the ^{37}Cl, D, and ^{13}C isotopologues of ClZnCH$_3$ have been conducted, enabling the first accurate structural determination of this molecule. The Zn—Cl and Zn—C bond distances are significantly shorter than those established by crystallography for ClZnCH$_2$CH$_3$ and ClZnEtTMEDA, which both have unusual solid-state structures. These data will help further benchmark computational studies of organozinc compounds, which are of fundamental interest in organic synthesis.

CRediT authorship contribution statement

M.A. Burton: Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **N. Tabassum:**

Investigation, Writing - original draft. **L.M. Ziurys:** Conceptualization, Methodology, Resources, Project administration, Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank Kyle Kilchenstein for his initial assistance. This work was supported by NSF grant CHE-1565765.

References

[1] Z. Rapporport, I. Marek, The Chemistry of Organozinc Compounds, John Wiley & Sons, England, 2006.
[2] A. Jõgi, U. Mäeorg, Molecules 6 (2001) 964–968.
[3] F. Zhou, C. Li, Nature Comm. 5 (2014) 4254.
[4] E. Negishi, A.O. King, N. Okukado, J. Org. Chem. 42 (1977) 1821.
[5] A.O. King, N. Okukado, E. Negishi, J. Chem. Soc., Chem. Commun. (1977) 683.
[6] N. Alonso, L.Z. Miller, J. de M. Muñoz, J. Alcázar, D.T. McQuade, Adv. Synth. Catal. 356 (2014) 3737–3741.
[7] B.A. Haag, Z.-G. Zhang, J.-S. Li, P. Knochel, Agnew. Chem. Int. Ed. 49 (2010) 9513–9516.
[8] I. Kumar, R. Kumar, U. Sharma, Synthesis 50 (2018) 2655–2677.
[9] X. Lin, D.L. Phillips, J. Org. Chem. 73 (2008) 3680–3688.
[10] G.T. de Jong, R. Visser, F.M. Bickelhaupt, J. Organomet. Chem. 691 (2006) 4341–4349.
[11] P.T. Moseley, H.M.M. Shearer, Chem. Commun. (1966) 876–877.
[12] P.T. Moseley, H.M.M. Shearer, J. Chem. Soc. Dalton Trans. (1973) 64–67.
[13] P.C. Andrews, C.L. Raston, B.W. Skelton, A.H. White, Organometallics 17 (1998) 779–782.
[14] A. Guerrero, D.L. Hughes, M. Bochmann, Organometallics 25 (2006) 1525–1527.
[15] Y. Nishida, N. Hosokawa, M. Murai, K. Takai, J. Am. Chem. Soc. 137 (2015) 114–117.
[16] M.A. Flory, A.J. Apponi, L.N. Zack, L.M. Ziurys, J. Am. Chem. Soc. 132 (2010) 17186–17192.
[17] M.P. Bucchino, J.P. Young, P.M. Sheridan, L.M. Ziurys, J. Phys. Chem. A 118 (2014) 11204–11210.
[18] J. Min, M.P. Bucchino, K.M. Kilchenstein, L.M. Ziurys, Chem. Phys. Lett. 646 (2016) 174–178.
[19] L.M. Ziurys, W.L. Barclay Jr., M.A. Anderson, D.A. Fletcher, J.W. Lamb, Rev. Sci. Instrum. 65 (1994) 1517–1522.
[20] C.H. Townes, A.L. Schawlow, Microwave Spectroscopy. Dover NY (1975).
[21] H.M. Pickett, J. Mol Spec. 148 (1991) 371–377.
[22] Z. Kisiel, J. Mol. Spectrosc. 218 (2003) 58–67.
[23] J.K.G. Watson, A. Roytburg, W. Ulrich, J. Mol. Spectrosc. 196 (1999) 102–119.
[24] J.L. Duncan, J. Mol. Spectrosc. 22 (1974) 225–235.
[25] E.D. Tenenbaum, M.A. Flory, R.L. Pulliam, L.M. Ziurys, J. Mol. Spectrosc. 244 (2007) 153–159.
[26] S. Elmoussaoui, N. El-Kork, M. Korek, Comput, Theor. Chem. 1090 (2016) 94–104.